

Recent announcements



Assignment 8 is corrected

https://mau.instructure.com/courses/22390/discussion_topics/447816

Due to other duties I have been late with the correction of the assignments and I do apologize for this! However, assignment 8 is now corrected and I will proceed with assignment 9 and late submissions/resubmissions of assignments 1-6 this week. Thanks for your patience! Best wishes, Jörgen

Posted on:

20 Apr 2026, 11:19



Deadline assignment 7

https://mau.instructure.com/courses/22390/discussion_topics/443715

Dear students, Please note that today is the last day to hand in assignment 7 (project plan). For those of you who already finished assignments 1-8, assignment 9 is now available. Have a nice weekend! Best wishes, Jörgen

Posted on:

20 Mar 2026, 17:22



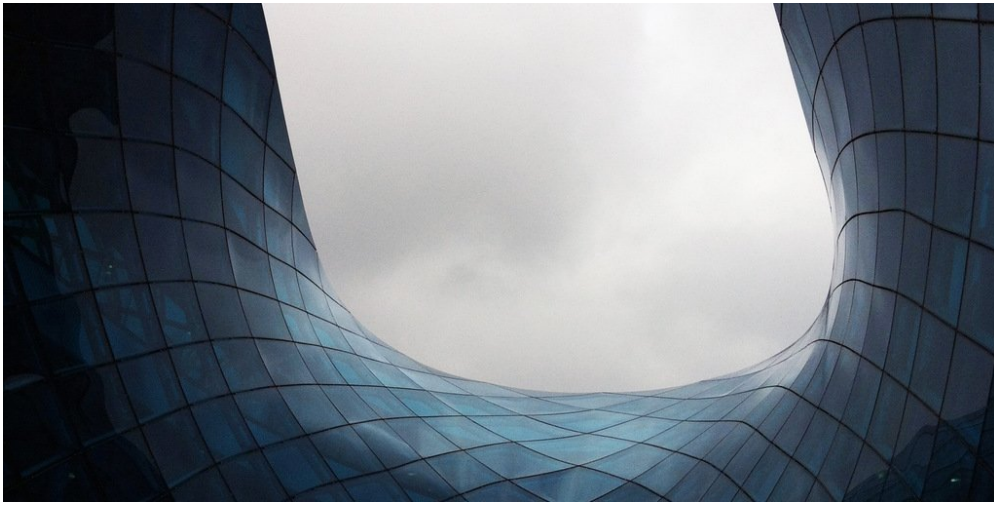
Assignment 7 is now available!

https://mau.instructure.com/courses/22390/discussion_topics/441330

Dear students, As you might have seen, assignment 7 is now available. This assignment is different in that you are asked to hand-in a 1-2 pages plan for the project. The idea is that you should choose one of three physics problems to study: I. Atomic structure II. Analysis of satellite data III. The carbon cycle and climate response You will find information about the projects on Canvas. The last hand-in date is Friday 20 March. Have a nice weekend. Best wishes, Jörgen

Posted on:

6 Mar 2026, 16:37



Purpose

The purpose of the course is that the student shall develop knowledge and basic skills to use numerical methods for modeling physical systems. The aim of the course is also that the student obtains skills to implement numerical methods in Python. See the [course syllabus](https://utbildningsinfo.mau.se/kurs/kursplan/en/1d94b583-3aa3-11ef-bd66-a3688dcf460c/20261)  [_for more details.](https://utbildningsinfo.mau.se/kurs/kursplan/en/1d94b583-3aa3-11ef-bd66-a3688dcf460c/20261)

Contacts

Jörgen Ekman  [_](https://mau.se/personer/jorgen.ekman/), senior lecturer in physics, course responsible

Namrata Nath  [_](https://mau.se/en/persons/namrata.nath/), doctoral student

Pedro Henrique Arantes Moya  [_](https://mau.se/personer/pedro.moya/), doctoral student

Kristina Olsson  [_](https://mau.se/personer/kristina.olsson/), student services assistant

Required software

The course 100 % online. To follow the course you are expected to have access to a personal computer with Python and preferably an integrated development environment (IDE) installed. As IDE we recommend Spyder. In addition, in some of the exercises, you are asked to perform screen recordings of various activities, and therefore you should also install a screen recording program suitable for the operating system on your personal computer.

Windows

Spyder can be downloaded here: <https://www.spyder-ide.org/>  [_](https://www.spyder-ide.org/).

Mac

Mac OS X comes with Python pre-installed. If you need or want a different Python version, it can be downloaded from <https://www.python.org>  [_](https://www.python.org/). Spyder can be

downloaded here: <https://www.spyder-ide.org/>  [\(https://www.spyder-ide.org/\)](https://www.spyder-ide.org/).

For Homebrew users: [Python](https://docs.brew.sh/Homebrew-and-Python)  [_\(https://docs.brew.sh/Homebrew-and-Python\)_](https://docs.brew.sh/Homebrew-and-Python).

Linux

Most Linux distributions come with Python pre-installed. If you need or want a different Python version, it can be downloaded from <https://www.python.org/>  [_\(https://www.python.org/\)_](https://www.python.org/). Spyder can be downloaded here: <https://www.spyder-ide.org/>  [_\(https://www.spyder-ide.org/\)_](https://www.spyder-ide.org/).

Lecture Notes

The Lecture Notes that you need for the course can be found here: [LectureNotes.pdf](#)

[_\(https://mau.instructure.com/courses/22390/files/3525066?wrap=1\)_](https://mau.instructure.com/courses/22390/files/3525066?wrap=1) 

[_\(https://mau.instructure.com/courses/22390/files/3525066/download?download_frd=1\)](https://mau.instructure.com/courses/22390/files/3525066/download?download_frd=1)

Files used in the lecture notes:

[FilesToLectureNotesExamples.zip](#) [_\(https://mau.instructure.com/courses/22390/files/3525085?wrap=1\)_](https://mau.instructure.com/courses/22390/files/3525085?wrap=1)  [_\(https://mau.instructure.com/courses/22390/files/3525085/download?download_frd=1\)](https://mau.instructure.com/courses/22390/files/3525085/download?download_frd=1)

Solutions to most of the exercises in the lecture notes:

[losningar_python_kap1_12.zip](#) [_\(https://mau.instructure.com/courses/22390/files/3525067?wrap=1\)_](https://mau.instructure.com/courses/22390/files/3525067?wrap=1)  [_\(https://mau.instructure.com/courses/22390/files/3525067/download?download_frd=1\)](https://mau.instructure.com/courses/22390/files/3525067/download?download_frd=1)

[losningar_python_kap13_20.zip](#) [_\(https://mau.instructure.com/courses/22390/files/3525068?wrap=1\)_](https://mau.instructure.com/courses/22390/files/3525068?wrap=1)  [_\(https://mau.instructure.com/courses/22390/files/3525068/download?download_frd=1\)](https://mau.instructure.com/courses/22390/files/3525068/download?download_frd=1)

Assignments

For assessment you are required to hand-in a total of nine (9) sets of exercises and a project report according to the table below. Each assignment description, available as a pdf document, consists of a set of exercises to be handed in and for many of the exercises also a physics background is given for those of you who are interested and want to learn more. The assignment descriptions are available through a link in the table below at least two weeks before the latest hand-in date.

Exercise sets are handed in on Canvas and we strongly encourage the use of LaTeX and the online tool [Overleaf](#)  [_\(https://www.overleaf.com/\)_](https://www.overleaf.com/). To get started with Overleaf, which is not difficult, please consult this [short manual](#)

[_\(https://mau.instructure.com/courses/22390/files/3506485/download?wrap=1\)_](https://mau.instructure.com/courses/22390/files/3506485/download?wrap=1) 

[_\(https://mau.instructure.com/courses/22390/files/3506485/download?download_frd=1\)_](https://mau.instructure.com/courses/22390/files/3506485/download?download_frd=1) . In this [quick guide](#) [_\(https://mau.instructure.com/courses/22390/files/3506481/download?wrap=1\)_](https://mau.instructure.com/courses/22390/files/3506481/download?wrap=1) 

[_\(https://mau.instructure.com/courses/22390/files/3506481/download?download_frd=1\)_](https://mau.instructure.com/courses/22390/files/3506481/download?download_frd=1) more information about type setting in LaTeX is found. In [LaTeX tricks and conventions](#)

[_\(https://mau.instructure.com/courses/22390/files/3506486?wrap=1\)_](https://mau.instructure.com/courses/22390/files/3506486?wrap=1) 

[_\(https://mau.instructure.com/courses/22390/files/3506486/download?download_frd=1\)](https://mau.instructure.com/courses/22390/files/3506486/download?download_frd=1)

[_\(https://mau.instructure.com/courses/22390/files/3506486?wrap=1\)_](https://mau.instructure.com/courses/22390/files/3506486?wrap=1) you can find some valuable

information when writing reports with LaTeX. A complete description of LaTeX can be found [here](https://mau.instructure.com/courses/22390/files/3506521?wrap=1) (<https://mau.instructure.com/courses/22390/files/3506521?wrap=1>)_ ↓
https://mau.instructure.com/courses/22390/files/3506521/download?download_frd=1) . Additional support is provided upon request.

Please use the following templates for hand-in exercises:

LaTeX (Overleaf): [Zip-file](https://mau.instructure.com/courses/22390/files/3506552?wrap=1) (<https://mau.instructure.com/courses/22390/files/3506552?wrap=1>)_ ↓
https://mau.instructure.com/courses/22390/files/3506552/download?download_frd=1) that you can upload to Overleaf to start write the report for the assignments (in Overleaf, select New Project and then Upload Project).

LaTeX (Other than Overleaf): [template](https://mau.instructure.com/courses/22390/files/3506555?wrap=1) (<https://mau.instructure.com/courses/22390/files/3506555?wrap=1>)_ ↓
https://mau.instructure.com/courses/22390/files/3506555/download?download_frd=1) and [university logotype](https://mau.instructure.com/courses/22390/files/3506554?wrap=1) (<https://mau.instructure.com/courses/22390/files/3506554?wrap=1>)_ ↓
https://mau.instructure.com/courses/22390/files/3506554/download?download_frd=1) .

MS Word: [template](https://mau.instructure.com/courses/22390/files/3534269?wrap=1) (<https://mau.instructure.com/courses/22390/files/3534269?wrap=1>)_ ↓
https://mau.instructure.com/courses/22390/files/3534269/download?download_frd=1)

The exercise sets are supposed to be handed as one single document. The answer to each exercise should in general follow this structure:

1. Define the problem/exercise to solved
2. Give the necessary, and well commented, Python code used to solve the problem/exercise
3. Give the result

As an example, please have a look at the file [Assignment_handin_example_2026.pdf](https://mau.instructure.com/courses/22390/files/3506553?wrap=1) (<https://mau.instructure.com/courses/22390/files/3506553?wrap=1>)_ ↓
https://mau.instructure.com/courses/22390/files/3506553/download?download_frd=1) . This is the output pdf file when processing the Overleaf zip-file above.

Please note: Solutions of exercise sets/assignments to be handed in for examination should be the result of your own work. You are not allowed to use generative AI tools when solving the exercise sets/assignments to be handed in for examination. Doing so will be reported as disciplinary misconduct and may result in disciplinary action.

Assignments 1-3, 1.5 credits

Latest hand-in date

1. [Introduction to Python](https://mau.instructure.com/courses/22390/pages/assignment-1-introduction-to-python)

(<https://mau.instructure.com/courses/22390/pages/assignment-1-introduction-to-python>)

Friday 30 January

2. [Programming with Python](https://mau.instructure.com/courses/22390/pages/assignment-2-programming-with-python)

(<https://mau.instructure.com/courses/22390/pages/assignment-2-programming-with-python>)

Friday 6 February

3. [Interpolation and Curve Fitting Methods](https://mau.instructure.com/courses/22390/pages/assignment-3-interpolation-and-curve-fitting-methods)

(<https://mau.instructure.com/courses/22390/pages/assignment-3-interpolation-and-curve-fitting-methods>)

Friday 13 February

Assignments 4-6, 1.5 credits

4. Integrals and Roots**(<https://mau.instructure.com/courses/22390/pages/assignment-4-integrals-and-roots>)**

Friday 20 February

5. Differential Equations I**(<https://mau.instructure.com/courses/22390/pages/assignment-5-differential-equations-i>)**

Friday 27 February

6. Differential Equations II**(<https://mau.instructure.com/courses/22390/pages/assignment-6-differential-equations-ii>)**

Friday 6 March

Assignments 7-9, 1.5 credits**7. Project Plan****(<https://mau.instructure.com/courses/22390/pages/assignment-7-project-plan>)**

Friday 20 March

8. Eigen-value Problems**(<https://mau.instructure.com/courses/22390/pages/assignment-8-eigen-value-problems>)**

Friday 27 March

9. Monte-Carlo Methods**(<https://mau.instructure.com/courses/22390/pages/assignment-9-monte-carlo-methods>)**

Friday 3 April

Project Report, 3.0 credits**Project report****(<https://mau.instructure.com/courses/22390/pages/project>)**

Friday 22 May

Course Literature

Compendia of theory and assignment descriptions are provided (see above). Suggested reference literature are given below and additional literature is chosen in consultation with the teacher/tutor depending on the direction of the project.

Reference Literature:

Newman, M. (2012). Computational Physics. U.S.: CreateSpace